

Jonathan the old

Jonathan, the 176 year old tortoise is considered to be the world's oldest living animal

Tomorrow's Tech

body are made up of various components, but the ones that we are going to consider are the chromosomes. These are super-coiled clumps of DNA (the template for life if you will) and contain all the possible data that is needed to describe you as a whole. These chromosomes need to be replicated identically in every cell in your body and are absolutely essential. So essential in fact, that the body has very advanced and reliable mechanisms to ensure that any cell with damaged DNA is destroyed immediately. The cell is forced to commit suicide, apoptosis according to biologists.

During normal cell division (mitosis in this case), chromosomes split exactly in half, ensuring that by the end of the process two daughter cells are formed which are identical to the parent. This is how most of the cells in our body are replenished. It may seem to be an ideal process, one in which cells can literally reproduce forever, but there is a catch as you will see.

Telomeres

At its tip, each chromosome has a protective "cap" called a telomere which protects it from breaking up, and also prevents it from fusing with another chromosome. This ensures ensuring that the DNA sample it carries remains undamaged. The telomere unfortunately, gets depleted every time the cell divides till finally, it can no longer protect the chromosomes. At this point, there is a chance that the chromosome gets damaged and the cell self-destructs to ensure that the DNA is not contaminated. This, in effect, is how ageing takes place.

Certain cells like the sperm and egg cells in our body produce an enzyme called "telomerase reverse transcriptase" (TERT), which increases the length of the telomere as and when required, potentially making the cell immortal. In fact, if any cell can produce this enzyme continuously, it can potentially live forever. Unfortunately for us, the only cells in our body that have managed to do this, other than the above mentioned egg and sperm cells, are cancer cells. If cells continue to replicate after the telomere is depleted, DNA contamination

can and does occur, this contamination usually results in a cancer cell, especially if TERT production is accidentally activated.



Telomeres seen at the tips of the chromosomes

The goal

To prevent ageing, or even slow it down, it is essential that your cells divide and replicate for the longest possible time. If all your healthy cells replicate forever, you are immortal. To do this, scientists are trying to figure out a way of ensuring that TERT production is optimised. This may seem like a simple enough task, but it is far more complicated than just injecting the enzyme into your body, each type of cell requires a different, but exceptionally precise quantity of TERT, at regular intervals. The only way to achieve this is to make sure that the cells are actually doing this on their own and thus, genetic modification seems to be the only way. To put it more simply, the goal of every single researcher who is trying to prevent ageing is just one thing, sustained rejuvenation.

The solution

As far as scientists can make it out, it is currently impossible to be immortal, there are just too many natural and artificial sources of harm that are continuously damaging your body. With the right techniques and proper care however, it is possible to live for at least 400-500 years or more. The average human lifespan in developed countries is about 70-80 years, with proper dietary care and exercise, this can be extended to a theoretical 130 years or more in ideal circumstances.



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